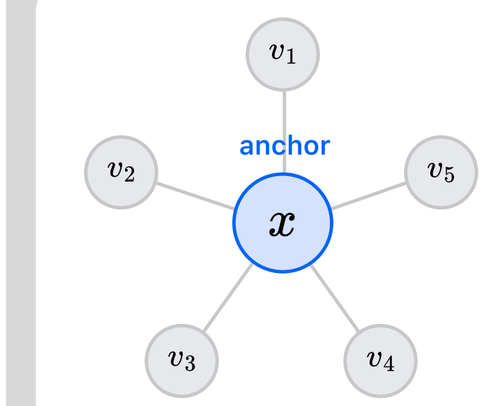
GRASP-RAG

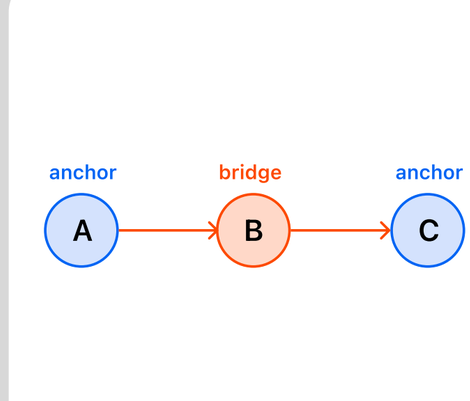
User Query

Candidate Entity Retrieval
top-20 nodes by semantic similarity

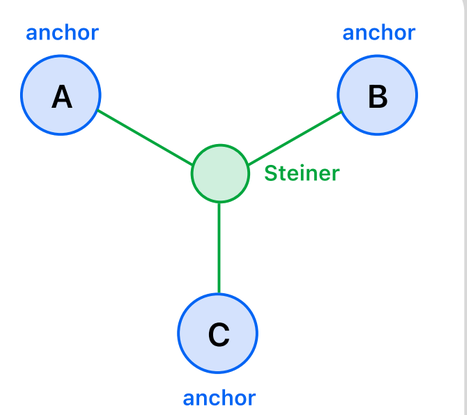
LLM Query Classifier
identifies query_type and anchor nodes



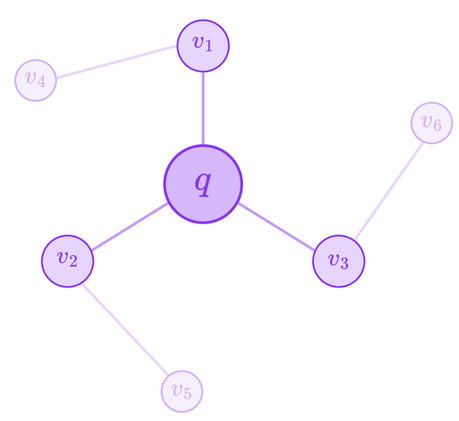
Ego-Network
single_entity



Shortest Path
bridge



Steiner Tree
multi_entity



Motif Cascade
broad / uncertain

Path-Guided Reasoning (PGR)
context mirrors graph topology
hop-chain / entity profile / comparison scaffold

LLM Answer Generation
GPT-4o-mini

Final Answer

Evaluation

Multi-Judge LLM Evaluation
forward + reverse ordering per judge

GPT-4o-mini

Claude Haiku

Gemini 2.0 Flash

Bias-Corrected Majority Vote
2/3 judges must agree in both orderings

WIN | **LOSS** | **UNDEFINED**

per dimension (5 criteria)